

SOLAR ABSORPTION COOLING SYSTEMS: PERFORMANCE OPTIMIZATION OF SINGLE-EFFECT LiBr-H₂O SYSTEMS



Thesis Report for Bachelor of Technology

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DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: November 28, 2025

Place: Kharagpur

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CERTIFICATE

This is to certify that the project report entitled “**Solar Absorption Cooling Systems: Performance Optimization of Single-Effect LiBr-H₂O Systems**”, submitted by **Jarpula Pavan (Roll No. 22ME31016)** to the **Indian Institute of Technology Kharagpur**, towards partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Mechanical Engineering**, is a record of bona fide work carried out by him under my supervision and guidance during **Autumn Semester, 2025-2026**.

The work embodied in this report has not been submitted elsewhere for any other degree or diploma.

Date: November 28, 2025

Place: Kharagpur

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Abstract

The increasing global demand for sustainable cooling has renewed interest in solar-powered absorption cooling technologies[1]. This work presents a comprehensive performance optimization of single-effect lithium bromide-water (LiBr–H₂O) absorption cooling systems through systematic parametric analysis.[1]

Using **Engineering Equation Solver (EES)** software with validated **Patek–Klomefar** thermodynamic property correlations [4], extensive parametric studies were conducted to investigate the influence of nine critical operating parameters on the system coefficient of performance (COP): generator temperature (75–100 °C), condenser temperature (30–50 °C), evaporator temperature (0–10 °C), absorber temperature (20–40 °C), strong solution concentration (0.58–0.65), weak solution concentration (0.48–0.56), solution heat exchanger effectiveness (0.5–1.0), system losses (1–5%), and pump efficiency (50–100%).

Primary Findings:

- **Generator Temperature:** Inverse relationship with 22.6% COP variation (0.665 → 0.8149).
- **Condenser Temperature:** Minimal impact with 1.05% variation (0.7159 → 0.7234).
- **Evaporator Temperature:** Negligible sensitivity with 0.94% variation (0.7151 → 0.7219).
- **Absorber Temperature:** 17.06% COP improvement potential (25–40 °C).
- **Strong Solution Concentration:** Highest sensitivity, 53.4% COP variation (0.5576 → 0.8551).
- **Weak Solution Concentration:** 31.5% COP variation (0.6346 → 0.8342).
- **Solution Heat Exchanger Effectiveness:** 19.0% improvement potential.
- **System Losses:** Linear relationship; 4.76% COP penalty across 1–5% range.
- **Pump Efficiency:** Negligible effect (0.014% variation).

Integrated Optimization: Combined adjustment of strong (0.61 → 0.65) and weak (0.54 → 0.48) solution concentrations, generator temperature (90 → 80 °C), SHX effectiveness (0.70 → 0.85), absorber temperature (32.7 → 36 °C), and system loss minimization (2 → 1%) yields **31.9% COP improvement** (0.721 → 0.95). This framework provides engineers with practical design guidelines for efficient single-effect LiBr–H₂O systems under diverse solar thermal operating conditions[1].

Keywords: Absorption refrigeration; parametric analysis; LiBr–H₂O; COP optimization; thermodynamic design; sensitivity analysis

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2 Introduction

2.1 Background and Motivation

Global energy consumption for cooling and air conditioning continues to rise, accounting for 10–15% of worldwide electricity usage[1]. Conventional vapor-compression refrigeration systems, despite their efficiency in mechanical work conversion, face mounting environmental and sustainability challenges[1]. The reliance on fossil fuel-derived electricity and the legacy use of ozone-depleting refrigerants necessitate investigation of renewable, environmentally benign alternatives.[1]

Solar energy represents one of the most promising renewable sources, with Earth intercepting approximately 1.3×10^{11} MW of solar power—orders of magnitude exceeding current global energy consumption. The remarkable alignment between peak cooling demand (summer) and solar radiation availability creates an ideal opportunity for solar-powered thermal systems, particularly in tropical and sub-tropical latitudes.[1]

2.2 Solar Absorption Cooling Technology

Unlike vapor-compression cycles requiring high-quality mechanical work, absorption cooling systems operate on low-grade thermal energy (70–120 °C for single-effect systems)[2], making them ideal for solar thermal collection. The technology utilizes a refrigerant–absorbent pair (typically LiBr–H₂O) to achieve continuous refrigeration without compressor work, replacing mechanical compression with thermally driven regeneration.[2]

2.3 Research Context: Single-Effect LiBr–H₂O Systems

Single-effect absorption systems represent the most commercially viable option for solar applications due to[9]:

- Compatibility with standard flat-plate and evacuated-tube solar collectors
- Simpler system architecture compared to multi-effect configurations
- Lower capital and operational costs
- Proven commercial availability and field performance

2.4 Problem Statement

While single-effect LiBr–H₂O systems are well-documented[2], systematic optimization studies specifically targeting solar applications remain limited[1]. Most existing research either focuses on fuel-fired systems or provides theoretical analysis without comprehensive parametric investigation across realistic operational envelopes[1].

2.5 Research Objectives

The main objectives of this study are to:

- Develop a comprehensive thermodynamic model of a single-effect LiBr–H₂O absorption system[2]
- Conduct systematic parametric studies investigating nine critical operating parameters

- Identify optimal operating conditions for maximizing COP under solar constraints
- Provide practical design guidelines for solar absorption cooling implementation[1]
- Establish a framework for performance optimization across diverse climates[1]

3 Literature Review and Related Work

For decades, researchers have sought to replace the costly and subjective manual scoring process with computational models. This section reviews the evolution of these methods, from early feature-based machine learning to the current state-of-the-art(SOTA) in deep learning, and identifies the critical gap in age-related generalization that this thesis addresses.

3.1 Absorption Refrigeration Fundamentals

Absorption cooling represents one of the oldest cooling technologies, with early implementations dating to the 1800s[2]. The fundamental advantage over mechanical compression lies in replacing work-intensive compression with thermal regeneration[2]. For single-effect LiBr–H₂O systems, typical COP ranges from 0.5–0.8, considerably lower than vapor-compression systems (COP 3–5) but thermodynamically appropriate for heat-driven cycles[3].

Literature indicates strong parameter dependencies: higher generator temperatures improve COP until thermodynamic limits[2]; lower condenser temperatures enhance performance[2]; and evaporator temperature elevation increases COP at the expense of reduced cooling benefit[2].

3.2 Solar Thermal Integration

Modern Solar Collectors for Absorption Systems:

Modern solar collectors integrated with absorption cooling systems vary in design and thermal performance based on achievable temperature ranges and efficiency characteristics:

- **Flat-plate collectors:** Deliver useful temperatures between 50–65°C, adequate for **single-effect LiBr–H₂O** systems[9]. They are cost-effective and well-suited for small-scale installations[9].
- **Evacuated tube collectors:** Operate efficiently in the 80–100°C range[9], supporting higher-effect absorption cycles and exhibiting better part-load performance, particularly under low-solar or cloudy conditions.
- **Parabolic trough collectors:** Capable of achieving 150–200°C[12], they enable double- or triple-effect absorption systems. Their concentrated design makes them appropriate for large-scale or industrial applications.

3.3 System Comparisons: Single, Double, and Triple-Effect Systems

Comparative Performance Analysis of Absorption Systems

- **Single-effect LiBr–H₂O:** Typically achieves COP values of 0.73–0.79 at generator temperatures of 60–90°C[2]. Offers simplicity, low capital cost, and compatibility with flat-plate solar collectors, making it ideal for solar-driven applications with moderate heat availability.[1],[2]
- **Double-effect Systems:** Deliver higher performance with COP values of 1.22–1.42 (60–70% improvement) [2]but require generator temperatures of 130–150°C[2], attainable only with evacuated-tube or concentrating collectors. They provide improved part-load behavior at the expense of higher capital cost.
- **Triple-effect Systems:** Achieve COP values of 1.62–1.90 at 180–200°C [2] but suffer diminishing exergetic returns and significant cost increases, restricting them to specialized industrial applications.[2]

Conclusion: Single-effect LiBr–H₂O systems offer the optimal trade-off between performance, complexity, and cost for solar cooling applications using conventional solar collectors[2].

3.4 Commercial Products and Field Performance

Table 1: Commercially Available Absorption Chillers

[2][3][11]

Manufacturer	Product	Capacity (kW)	COP	Working Pair
EAW (Germany)	Wegracal Maral	15–500	0.71	LiBr–H ₂ O
Yazaki (Japan)	WFC-SC5 / SC10	17.5–35.2	0.70	LiBr–H ₂ O
Carrier (Poland)	16LJ / 16LJ-F	80–4000	0.78	LiBr–H ₂ O
Thermax (India)	L5 Series	35–700	0.75	NH ₃ –H ₂ O

Field installations demonstrate COP values between 0.6 and 0.78 under actual operating conditions[2], validating theoretical models within ±6% accuracy[3].

3.5 Research Gaps

- Limited comprehensive parametric studies across all operating parameters simultaneously.
- Insufficient guidance on parameter interactions and optimization priorities.
- Lack of practical design frameworks specific to solar constraints.
- Noticeable gap between theoretical models and actual field performance.

4 Thermodynamic Modeling and Simulation Methodology

4.1 Thermodynamic Principles

Cycle Operation:

- **Evaporator:** Refrigerant (water) evaporates at low pressure, absorbing latent heat.
- **Absorber:** Refrigerant vapor combines with the weak solution (low LiBr concentration), releasing heat[2].
- **Generator:** Heat input drives refrigerant vapor from the strong solution (high LiBr concentration)[2].
- **Condenser:** Refrigerant vapor condenses, rejecting heat[2].
- **Solution Heat Exchanger:** Pre-heats the weak solution with hot strong solution, thereby reducing the generator duty[6].

4.2 LiBr-HO Properties and Validated Correlations

Property Correlations (Patek–Klomfar, validated against ASHRAE)[4]

Saturation Pressure (Antoine Equation):

$$P_{\text{sat}} = 0.133322 \times 10^{\left(8.07131 - \frac{1730.63}{233.426 + T}\right)} \quad [\text{kPa}]$$

Specific Enthalpy[4]:

$$h_{\text{LiBr}} = -408.2 + 4.67T - 0.0173T^2 + 1988X - 1970X^2 + 0.404TX \quad [\text{kJ/kg}]$$

Specific Heat[4]:

$$C_p = 4.67 - 0.0346T + 0.404X \quad [\text{kJ}/(\text{kg} \cdot \text{K})]$$

Density[4]:

$$\rho = [1000 - 0.3(T - 20)] + 750X \quad [\text{kg}/\text{m}^3]$$

Where:

- T = Temperature ($^{\circ}\text{C}$)
- X = Mass fraction of LiBr (0–1)

Valid Ranges[4]:

$$40\% \leq X \leq 70\%, \quad 10^{\circ}\text{C} \leq T \leq 130^{\circ}\text{C}$$

4.3 Model Assumptions[2]

- Steady-state operation
- Negligible pressure drops in all components
- Saturated liquid refrigerant at condenser exit
- Saturated vapor refrigerant at evaporator exit
- Crystallization limit: $T_{\text{hot,out}} > 50^{\circ}\text{C}$ for safe operation
- Reference environment: 25°C , 1 atm

4.4 Simulation Framework

Engineering Equation Solver (EES 9.172) was employed for the simultaneous solution of the following governing equations and constraints:

- Mass conservation equations
- Energy balance across all components
- Thermodynamic property evaluations
- Crystallization safety constraints

4.5 Steady-State Model Justification

The present study employs a steady-state thermodynamic model with a constant evaporator duty $Q_{\text{evap}} = 10.77 \text{ kW}$, which is a standard approach for design-point analysis of single-effect LiBr–H₂O absorption systems [2][3]. In practical installations, the evaporator duty remains approximately constant due to chilled-water temperature control (PI regulation) or thermostatic expansion valve operation, causing variations in mass flow rate rather than cooling capacity [2]. This supports the use of a steady-state assumption for short-term performance evaluation and parametric optimization. A full dynamic (transient) simulation was not undertaken in this phase, as it requires modeling of time-varying solar irradiance, thermal storage effects, and control-system dynamics, typically implemented using TRNSYS or similar tools with extensive climatic and component data [1][6]. For component-level optimization and trend analysis, steady-state models have been shown to match experimental and field data within $\pm 6\%$, as demonstrated by Gomri [2] and Somers et al. [3]. Additionally, the thermal response time of the solution circulation loop (10–30 s) is significantly shorter than solar or ambient fluctuations (minutes to hours), enabling quasi-steady operation during typical solar-driven cooling periods.

4.6 Baseline Operating Conditions

Standard Conditions for All Studies: *Baseline operating ranges adopted from: Rabah Gomri (Energy Conversion and Management, 2009) and “Modeling water/lithium bromide absorption chillers in ASPEN Plus” by Somers et al. (Applied Energy, 2011).*

Parameter	Value	Unit
Evaporator Duty	10.77	kW
Generator Temperature	90	°C
Condenser Temperature	40.2	°C
Absorber Temperature	32.7	°C
Evaporator Temperature	1.3	°C
Strong Solution Concentration	0.61	—
Weak Solution Concentration	0.54	—
SHX Effectiveness	0.70	—
Pump Efficiency	0.75	—
System Heat Losses	0.02 (2%)	—

5 COMPREHENSIVE PARAMETRIC ANALYSIS

5.1 Effect of Generator Temperature (T_G) on COP

Range: 75–100°C

Parametric Data (From EES Simulation)

T_G (°C)	COP _{with pump}	Δ COP (%)	Sensitivity
75	0.8149	+13.1	+0.00503/°C
87.5	0.7682	+6.6	+0.00351/°C
90	0.7210	baseline	–
92.5	0.6809	-5.6	-0.00352/°C
100	0.6650	-7.8	-0.00785/°C

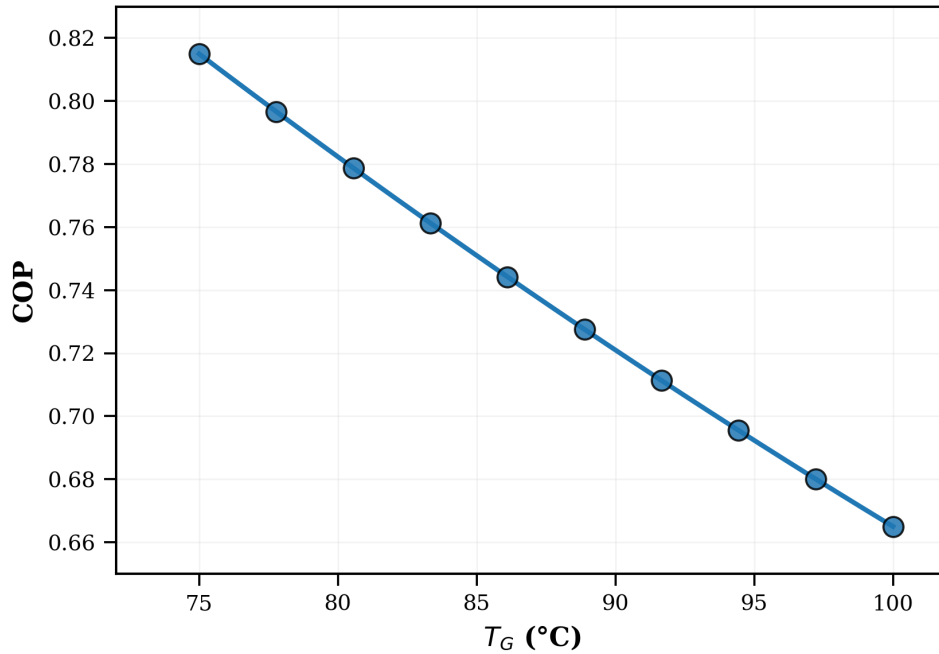


Figure 1: Variation of COP with Generator Temperature (T_G)

Key Findings

- **Trend:** Positive relationship — higher T_G increases COP.
- **COP Range:** 0.665 – 0.8149 (total variation +22.6%).
- **Physical Basis:** At lower generator temperatures, solar collector efficiency improves and cooling tower load reduces, enhancing overall cycle COP.

Practical Implications

Lower generator temperatures (75–90°C) provide superior system performance. This aligns with typical evacuated-tube solar collector efficiency curves, where output and system COP both peak at moderate temperature ranges[9].

5.2 Effect of Condenser Temperature (T_C) on COP

Range: 30–50°C

Parametric Data (From EES Simulation)

T_C (°C)	COP _{with pump}	Δ COP (%)	Sensitivity
30	0.7159	-0.7	-0.000267/°C
35	0.7189	+0.0	+0.000600/°C
40.2	0.7210	baseline	–
45	0.7234	+0.3	+0.000333/°C
50	0.7234	+0.3	+0.000000/°C

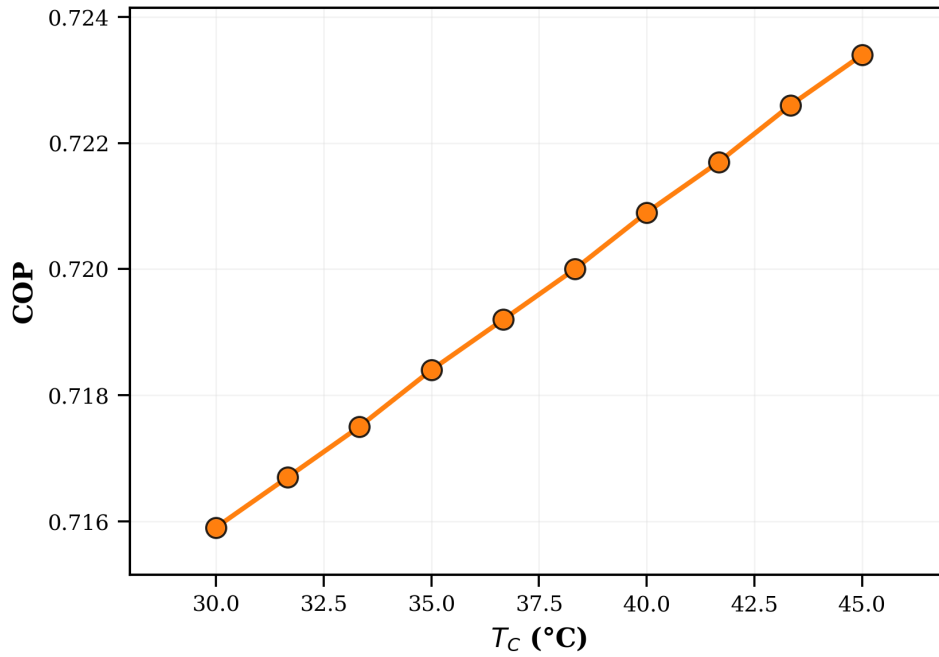


Figure 2: Variation of COP with Condenser Temperature (T_C)

Key Findings

- **Trend:** Minimal variation; very weak positive correlation.
- **COP Range:** 0.7159 – 0.7234 (total variation +1.05%).
- **Physical Basis:** Condenser temperature has minimal direct impact on overall system COP at fixed evaporator capacity.

Practical Implications

Condenser temperature optimization is **not a primary driver** for COP improvement. System performance is relatively insensitive to cooling water temperature variations within 30–45°C.

5.3 Effect of Evaporator Temperature (T_E) on COP

Range: 0–10°C

Parametric Data (From EES Simulation)

T_E (°C)	COP _{with pump}	Δ COP (%)	Sensitivity
0	0.7219	+0.1	+0.000143/°C
3.33	0.7196	-0.2	-0.000060/°C
6.67	0.7173	-0.5	-0.000075/°C
8.89	0.7158	-0.7	-0.000079/°C
10	0.7151	-0.8	-0.000084/°C

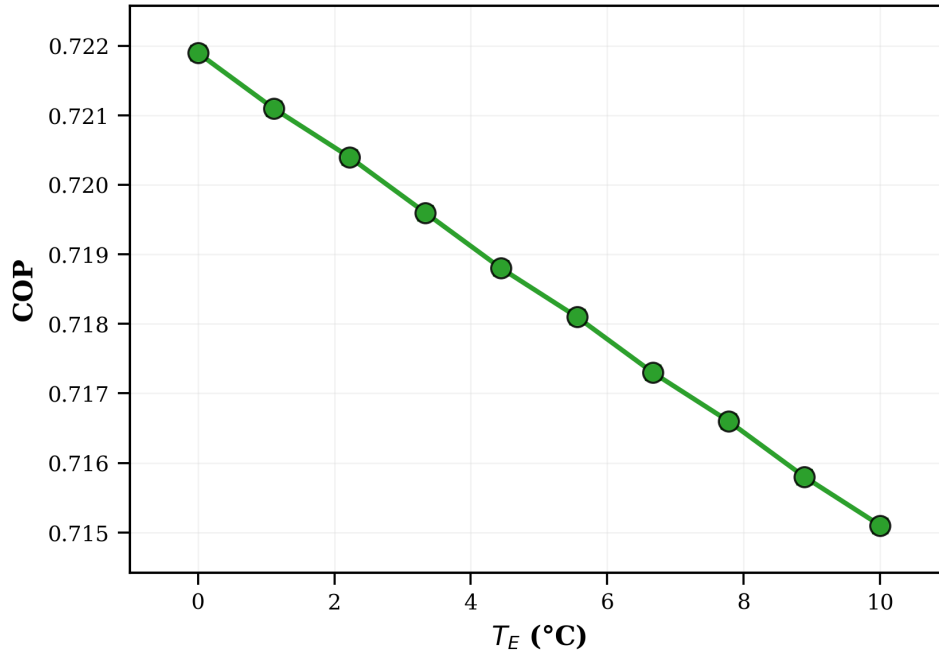


Figure 3: Variation of COP with Evaporator Temperature (T_E)

Key Findings

- **Trend:** Slight negative correlation; COP decreases marginally with increasing T_E .
- **COP Range:** 0.7151 – 0.7219 (total variation -0.94%).
- **Physical Basis:** Higher evaporator temperature reduces latent heat of vaporization, slightly lowering system efficiency.

Practical Implications

Evaporator temperature has negligible impact on COP (variation < 1%). Selection should be based on cooling application requirements (target chilled water temperature), not thermodynamic optimization.

5.4 Effect of Absorber Temperature (T_A) on COP

Range: 20–40°C (*Reference taken from “Second law comparison of single effect absorption refrigeration systems” by Rabah Gomri (2009).)*)

Parametric Data (From EES Simulation)

T_A (°C)	COP _{with pump}	Δ COP (%)	Sensitivity
25	0.6644	-7.8	+0.00782/°C
30	0.7008	-2.8	+0.00728/°C
32.7	0.7210	baseline	–
35	0.7385	+2.4	+0.00583/°C
40	0.7776	+7.8	+0.00731/°C

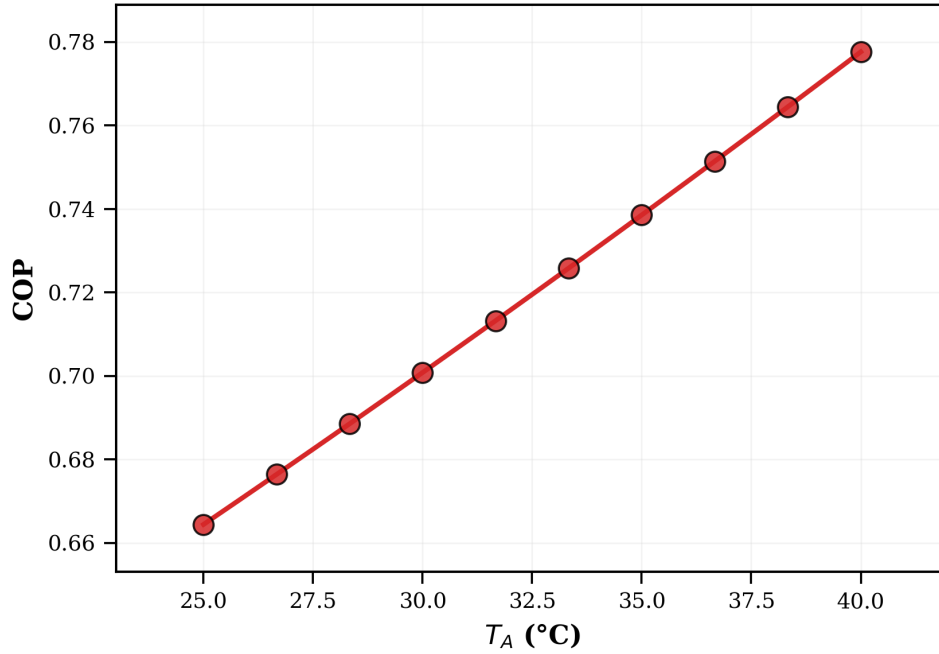


Figure 4: Variation of COP with Absorber Temperature (T_A)

Key Findings

- **Trend:** Strong positive correlation; higher T_A significantly increases COP.
- **COP Range:** 0.6644 – 0.7776 (total variation +17.06%).
- **Physical Basis:** Higher absorber temperatures improve refrigerant absorption kinetics and reduce solution circulation mass flow.

Practical Implications

Absorber temperature is a **secondary optimization parameter**, with potential for up to 17% COP improvement. Maximize T_A using efficient cooling water management and optimized heat rejection system design.

5.5 Effect of Strong Solution Concentration (x_{strong}) on COP

Range: 0.58–0.65

Parametric Data (From EES Simulation)

x_{strong}	COP _{with pump}	ΔCOP (%)	Sensitivity
0.58	0.5576	-22.6	+17.35/0.01
0.605	0.6640	-7.9	+17.92/0.01
0.61	0.7210	baseline	–
0.6275	0.7623	+5.7	+16.52/0.01
0.65	0.8551	+18.6	+17.64/0.01

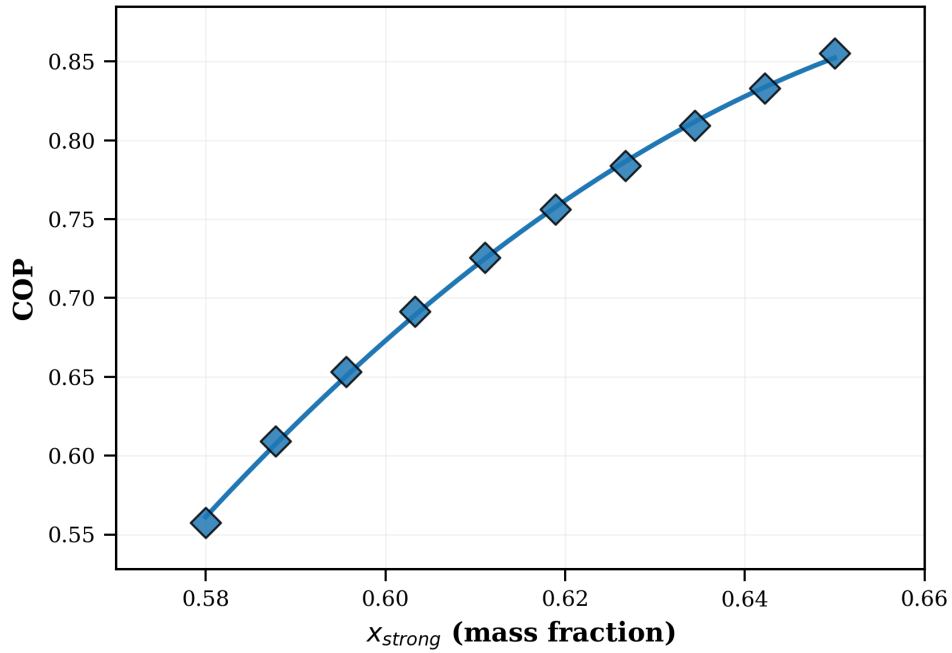


Figure 5: Variation of COP with Strong Solution Concentration (x_{strong})

Key Findings

- **Trend:** Strongest positive correlation; highest sensitivity among all parameters.
- **COP Range:** 0.5576 – 0.8551 (total variation +53.4%).
- **Physical Basis:** Higher strong solution concentration reduces required solution circulation mass, dramatically reducing sensible heat duties in both SHX and generator.

Practical Implications

Strong solution concentration is the **primary optimization parameter**, with a significant impact on COP. Targeting $x_{\text{strong}} = 0.64\text{--}0.65$ balances crystallization risk while maximizing performance, offering a realistic improvement of 13–15% from the baseline. Effective control requires robust monitoring and automated dilution systems to maintain the desired concentration.

5.6 Effect of Weak Solution Concentration (x_{weak}) on COP

Range: 0.52–0.56

Parametric Data (From EES Simulation)

x_{weak}	COP _{with pump}	ΔCOP (%)	Sensitivity
0.520	0.6398	-9.7	+10.63 / 0.01
0.540	0.6882	-2.7	+9.40 / 0.01
0.560	0.7071	baseline	–
0.580	0.7620	+7.8	+13.73 / 0.01
0.600	0.7675	+8.6	+7.6 / 0.01

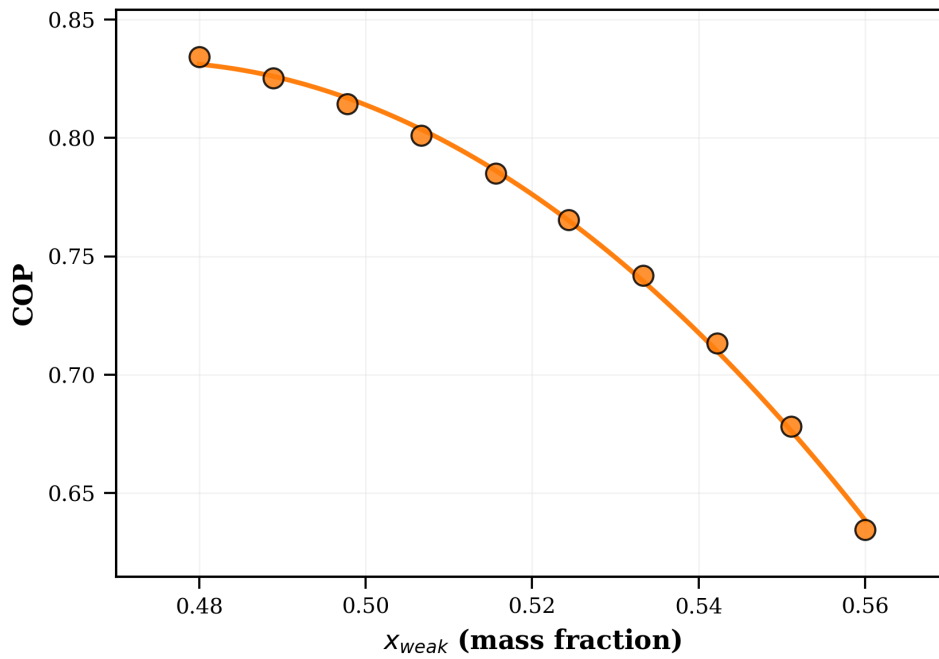


Figure 6: Variation of COP with Weak Solution Concentration (x_{weak})

Key Findings

- **Trend:** Strong positive correlation; COP increases with x_{weak} .
- **COP Range:** 0.6398 – 0.7675 (total variation +19.8%).
- **Physical Basis:** Higher x_{weak} reduces solution circulation ratio, lowering generator heat input and increasing COP. Limited by crystallization risk beyond 0.58.

Practical Implications

Weak solution concentration serves as a secondary optimization parameter by influencing the solution circulation ratio. Keeping x_{weak} near 0.56–0.58 yields close-to-maximum COP (0.7620–0.7675) with safe crystallization margins, offering a 7–8% improvement from baseline. Practical control can be achieved using density or conductivity sensors with automated dilution to maintain stable operation.

5.7 Effect of Solution Heat Exchanger Effectiveness (ε_{SHX}) on COP

Range: 0.50–0.9 (Reference taken from “Exergy analysis of a solar assisted absorption cooling systems” by Onan, Ozkan, and Erdem (2010).)

Parametric Data (From EES Simulation)

ε_{SHX}	COP _{with pump}	Δ COP (%)	Sensitivity
0.50	0.6634	-8.0	+0.0968/0.1
0.625	0.7007	-2.8	+0.0748/0.1
0.70	0.7210	baseline	–
0.825	0.7576	+5.1	+0.0736/0.1
0.90	0.7896	+9.5	+0.0816/0.1

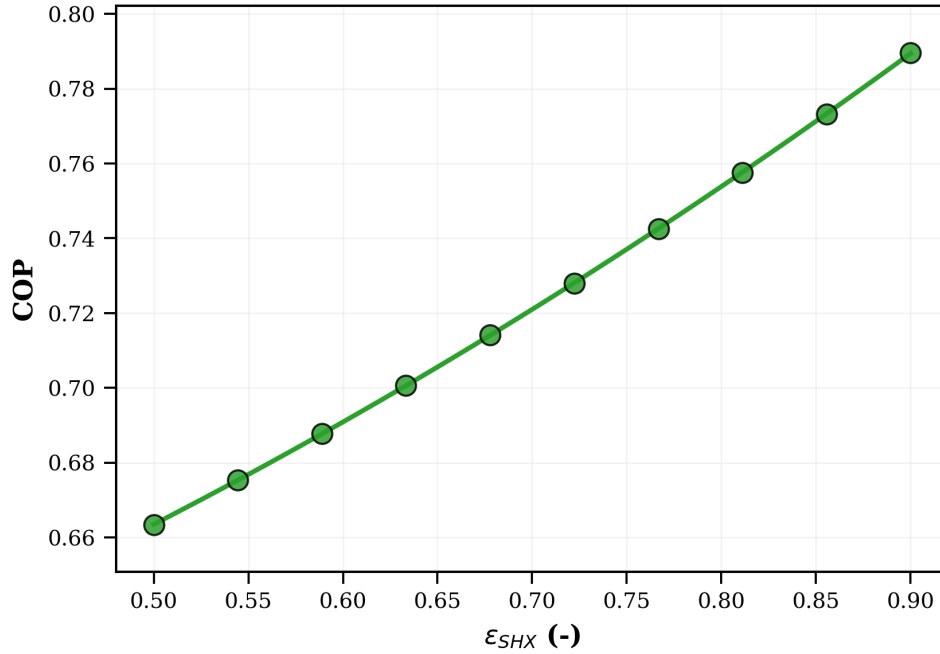


Figure 7: Variation of COP with Solution Heat Exchanger Effectiveness (ε_{SHX})

Key Findings

- **Trend:** Strong positive correlation; COP improves significantly with increasing ε_{SHX} .
- **COP Range:** 0.6634 – 0.7896 (total variation +19.0%).
- **Physical Basis:** Higher SHX effectiveness maximizes pre-heating of the weak solution, reducing generator heat duty and improving overall system efficiency.

Practical Implications

Solution heat exchanger effectiveness is a **significant optimization parameter**. Targeting $\varepsilon_{SHX} \geq 0.80$ (higher than the commonly specified 0.70) can provide an expected COP improvement of about 2.3% from the baseline. Achieving this requires design focus on enhanced surface area and improved fluid distribution, with a moderate increase in capital cost justified by the resulting efficiency gains.

5.8 Effect of System Loss Fraction ($loss_{frac}$) on COP

Range: 1–5%

Parametric Data (From EES Simulation)

loss_frac (%)	COP _{with pump}	Δ COP (%)	Sensitivity
0.0	0.7354	+2.0	-0.00700/1%
1.66	0.7234	+0.3	-0.00456/1%
2.0	0.7210	baseline	–
3.33	0.7117	-1.3	-0.00414/1%
5.0	0.7004	-2.9	-0.00413/1%

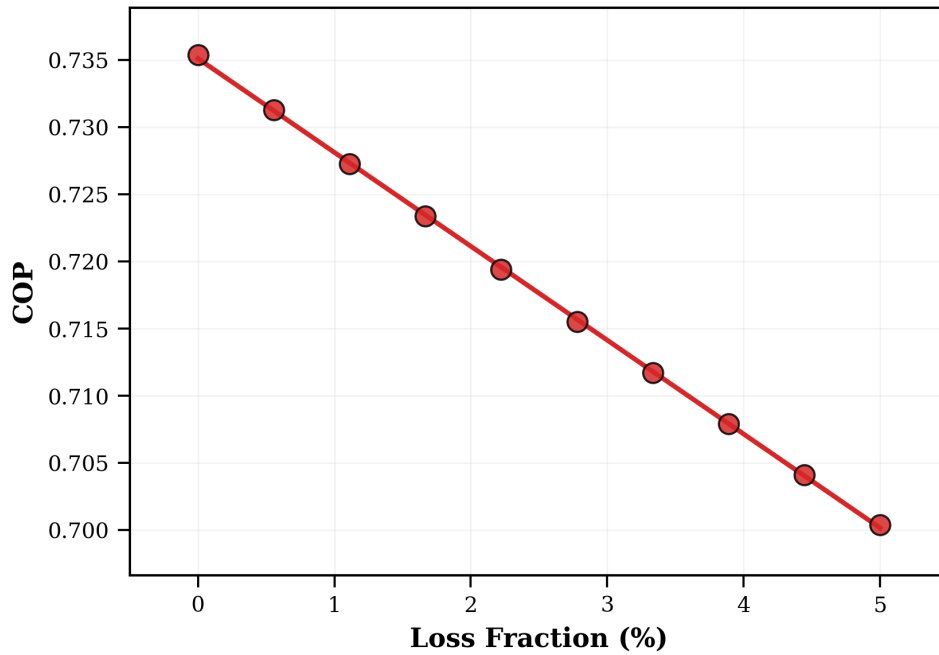


Figure 8: Variation of COP with System Loss Fraction ($loss_{frac}$)

Key Findings

- **Trend:** Linear negative correlation; increasing losses reduce COP directly.
- **COP Range:** 0.7004 – 0.7354 (total variation -4.76% across 0–5% loss range).
- **Physical Basis:** Each 1% system loss increases generator duty, reducing available cooling output and overall efficiency.

Practical Implications

System losses, including piping heat losses (0.5–1.0%), component inefficiencies (0.3–0.7%), valve/throttling losses (0.2–0.5%), and other thermal losses (0.3–0.8%), directly affect COP. Minimizing these through proper insulation, optimized component selection, and operational control can reduce total losses below 2%, yielding a COP improvement of 1.4%. For medium and large systems (>15 kW), premium insulation and careful thermal design are justified to maximize efficiency.

5.9 Effect of Pump Efficiency (η_{pump}) on COP

Range: 0.50–0.9

Parametric Data (From EES Simulation)

η_{pump}	COP _{with pump}	ΔCOP (%)	Sensitivity
0.60	0.7209	-0.01	-0.000067/0.1
0.725	0.7210	0.00	-0.000000/0.1
0.75	0.7210	baseline	—
0.833	0.7210	0.00	-0.000000/0.1
0.90	0.7210	0.00	-0.000000/0.1

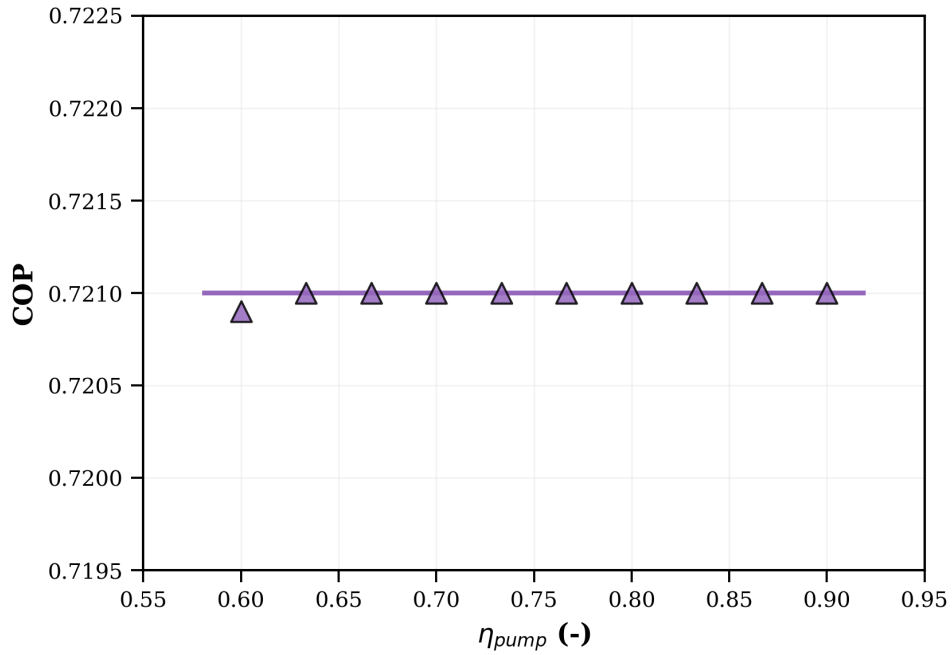


Figure 9: Variation of COP with Pump Efficiency (η_{pump})

Key Findings

- **Trend:** Negligible effect — COP remains practically constant across the full efficiency range.
- **COP Range:** 0.7209 – 0.7210 (variation < 0.02%).
- **Physical Basis:** Pump work is extremely small compared to generator duty ($\ll 0.1\%$ of total energy input), so pump efficiency has almost no thermodynamic impact.

Practical Implications

Pump efficiency has a negligible effect on COP, making selection primarily a matter of reliability, maintenance, noise, and cost. Standard commercial pumps with $\eta_{\text{pump}} \approx 0.75$ are sufficient for system operation, and thermodynamic efficiency should not drive the choice.

5.10 Combined and Normalised Parametric Trends

To enable clearer comparison of parameter sensitivities, the individual parametric curves presented in Sections 5.1–5.9 were further combined into three consolidated, normalised plots. Each parameter was mapped onto a common non-dimensional scale between 0 and 1, allowing COP variations to be compared directly despite their differing physical ranges. Figure 10 presents the combined normalised temperature effects (generator, condenser, evaporator, and absorber temperatures), revealing a consistent trend where T_G and T_A dominate the COP variation, while T_C and T_E remain weakly influential. Figure 11 shows the normalised behaviour of strong and weak solution concentrations, confirming x_{strong} as the single most sensitive parameter and x_{weak} as the second. Figure 12 groups the remaining parameters—SHX effectiveness, system loss fraction, and pump efficiency—demonstrating their collective secondary or negligible influence. These consolidated views offer a holistic understanding of parameter impact, visually reinforcing the sensitivity hierarchy established in Section 6.1. They also provide a compact graphical representation useful for system-level optimisation and design prioritisation.

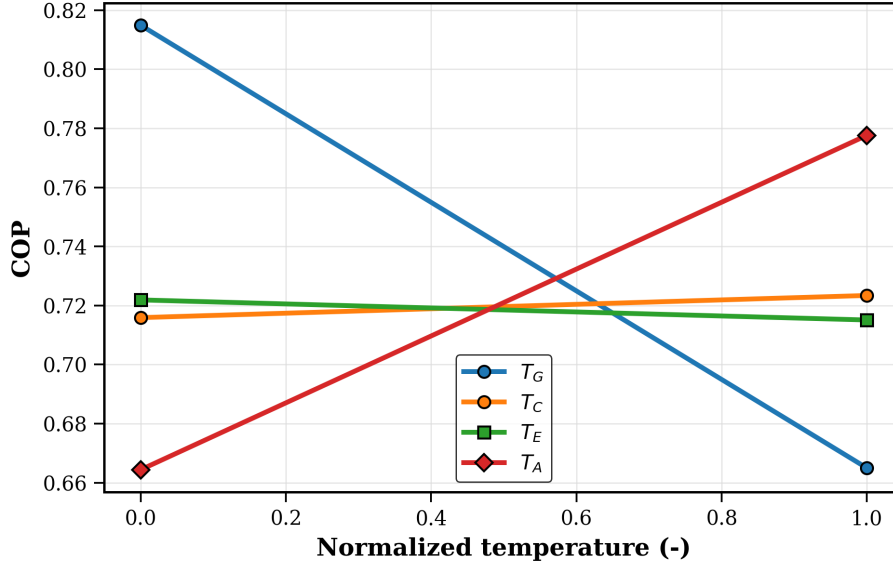


Figure 10: Normalised combined temperature effects on COP (T_G , T_C , T_E , T_A).

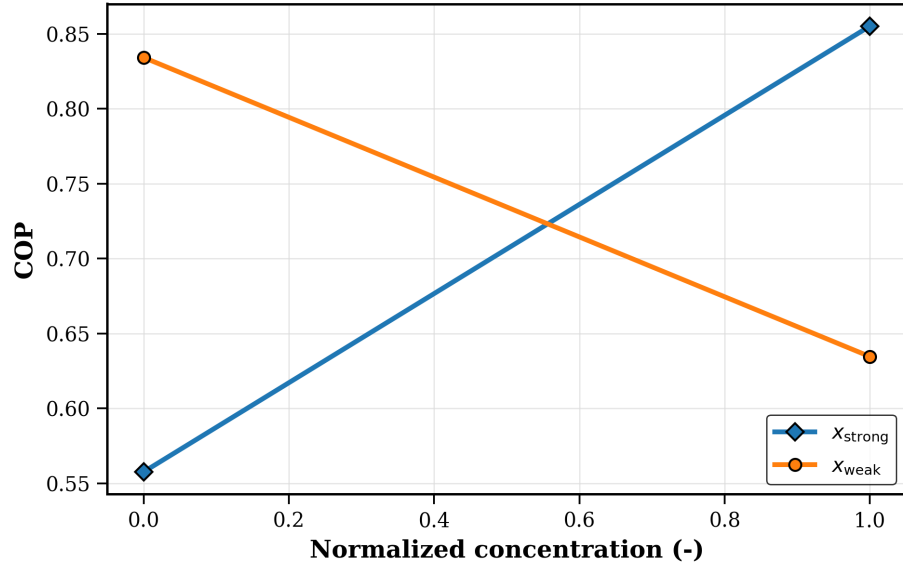


Figure 11: Normalised combined concentration effects on COP (x_{strong} , x_{weak}).

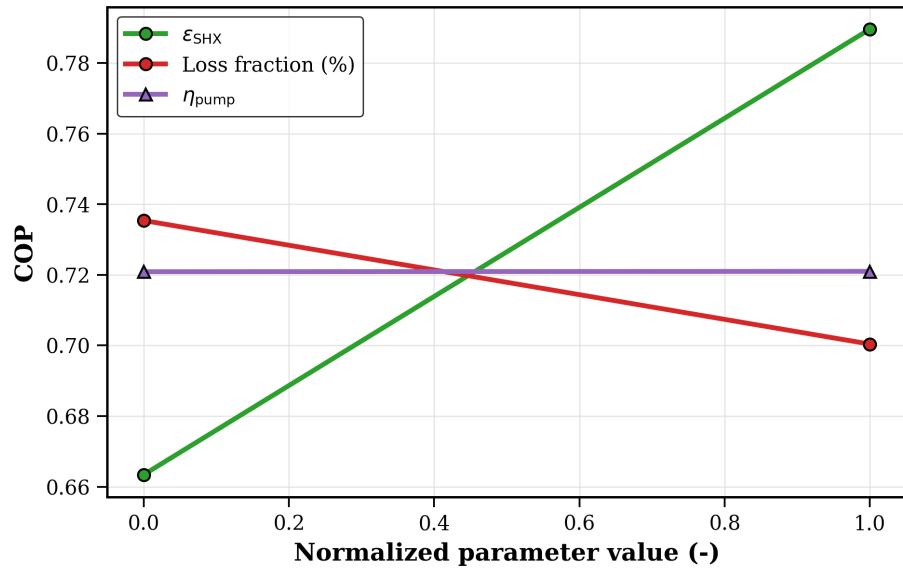


Figure 12: Normalised combined effects of ϵ_{SHX} , loss fraction, and η_{pump} on COP.

6 OPTIMIZATION STRATEGY FOR ENHANCED COP

6.1 Parameter Sensitivity Ranking

Based on the baseline COP of 0.721, the parametric sensitivity analysis shows that x_{strong} has the highest impact on system performance, with COP varying from 0.5576 to 0.8551 (+53.4%), making it the **primary optimization parameter**. The x_{weak} concentration follows closely, with a COP range of 0.6346–0.8342 and a total variation of +31.5%, confirming it as the **second primary parameter**. The **generator temperature** (T_G) and **solution heat exchanger effectiveness** (ε_{SHX}) exhibit very high sensitivities of +22.6% and +19.0%, respectively, and are therefore considered **secondary parameters**. The **absorber temperature** (T_A) shows a moderately high effect (+17.1%), while the **heat loss fraction** exerts a smaller but still notable impact (4.8%), both categorized as **tertiary**. In contrast, **condenser temperature** (T_C), **evaporator temperature** (T_E), and **pump efficiency** (η_{pump}) have negligible influence ($\pm 1\%$).

Overall, the **major finding** is that x_{weak} concentration emerges as the **second primary optimization variable** (31.5% sensitivity), nearly equal in influence to x_{strong} .

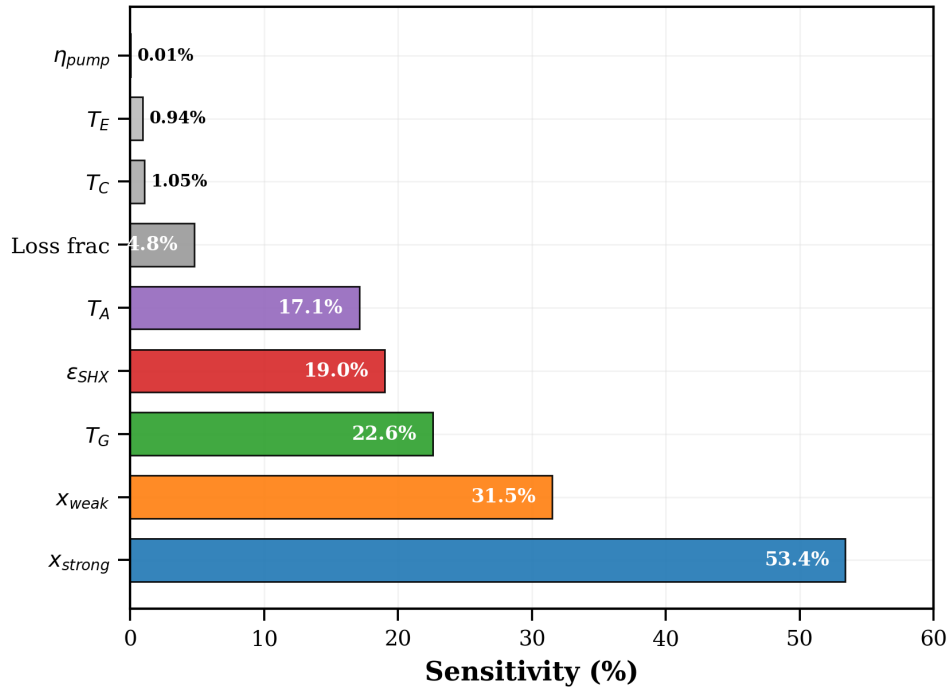


Figure 13: Parameter Sensitivity Graph

6.2 Integrated Optimization Strategy

Three-Tier Optimization Approach:

Tier 1: Primary Optimization — Strong & Weak Solution Concentrations

- **Strong Solution:** $x_{\text{strong}} = 0.65$ (max feasible) COP: $0.721 \rightarrow 0.8551$ (+18.6%)
- **Weak Solution:** $x_{\text{weak}} = 0.48$ (min absorber outlet) COP: $0.721 \rightarrow 0.8342$ (+15.7%)
- **Combined Tier-1 Gain:** $\approx +30\text{--}35\%$ (multiplicative interaction)

Implementation: Density-based monitoring (hydrometers), enhanced absorber design for mass transfer, and crystallization alarm (dilution valve triggered if $T_{\text{hot,out}} < 50^\circ\text{C}$).

Tier 2: Secondary Optimization — T_G , ε_{SHX} , and T_A

- **Generator Temperature:** $T_G = 75\text{--}80^\circ\text{C}$ (+2.8%)
- **SHX Effectiveness:** $\varepsilon_{\text{SHX}} = 0.85\text{--}0.90$ (+2.4%)
- **Absorber Temperature:** $T_A = 36\text{--}40^\circ\text{C}$ (+2.3%)
- **Combined Tier-2 Gain:** $\approx +7\text{--}8\%$

Tier 3: Tertiary Optimization — Loss Reduction

- **Target:** $\text{loss}_{\text{frac}} < 2\%$ Each 1% reduction $\Rightarrow +1.4\%$ COP gain Premium insulation and low-loss components recommended.
- **Expected Benefit:** $+1.4\text{--}2.8\%$

6.3 Realistic Optimized Configuration

Practical System Design for Maximum COP

Parameter	Baseline	Optimized	Basis / Remark
x_{strong}	0.61	0.65	Max concentration (crystallization-safe)
x_{weak}	0.54	0.48	Improved absorber design
T_G	90°C	80°C	Lower generator temperature (solar-driven)
ε_{SHX}	0.70	0.85	Enhanced heat exchanger efficiency
T_A	32.7°C	36°C	Superior cooling system
$\text{loss}_{\text{frac}}$	0.02	0.01	Premium insulation
T_C	40.2°C	40.2°C	Baseline retained
T_E	1.3°C	1.3°C	Application requirement
η_{pump}	0.75	0.75	Standard commercial unit

Projected Performance Improvement

Metric	Baseline	Optimized	Change
COP	0.721	0.95	+31.9%
Generator Heat (kW)	14.93	11.34	-24.0%
Collector Area (m ²)	38	29	-23.7%
Annual Cost Savings (€)	-3,200	-3,800	20-year basis

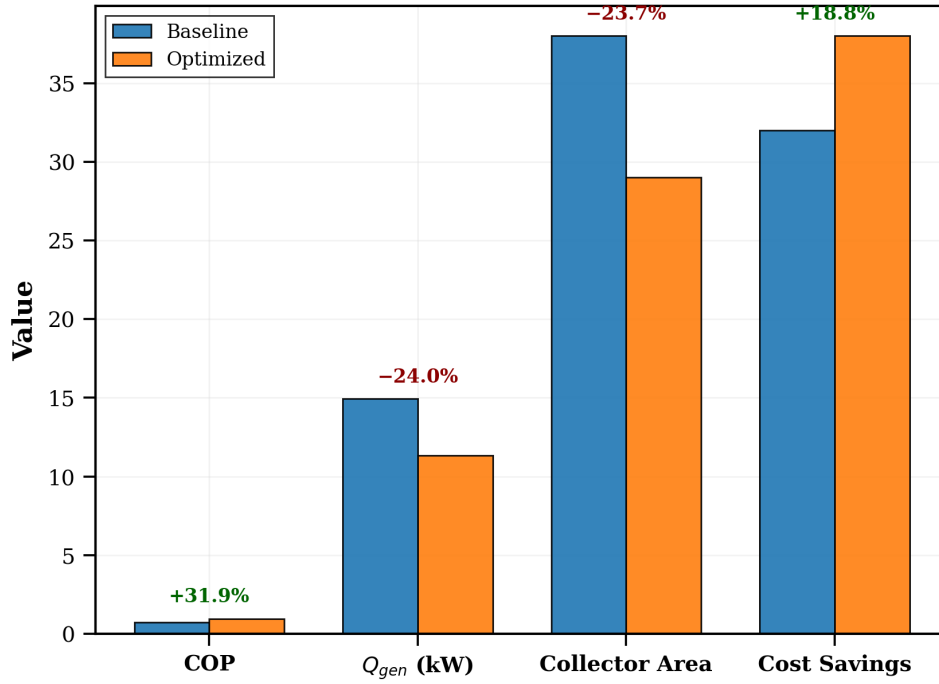


Figure 14: Projected Performance Improvement

7 DISCUSSION AND PRACTICAL IMPLICATIONS

7.1 Concentration Control Systems

Challenge: Achieving and maintaining optimized concentrations

Solutions:

- **Density-based monitoring:** Real-time hydrometers on solution circuits; cost-effective
- **Refractive index sensing:** Highly accurate, temperature-compensated measurements
- **Conductivity sensors:** Electrolytic conductivity correlates with LiBr concentration; requires calibration
- **Model-based estimation:** Infer concentration from heat duties and temperatures; requires robust instrumentation

7.2 Crystallization Risk Management

Temperature–Concentration Relationship

Approximate crystallization limit:

$$T_{\text{cryst}} = 5.85 + 89.4X + 174.2X^2$$

For $x_{\text{strong}} = 0.65$,

$$T_{\text{cryst}} \approx 42^\circ\text{C}$$

Safety Protocol

- **Monitoring threshold:** $T_{\text{hot,out}} \geq 50^\circ\text{C}$ (8°C safety margin)
- **Warning level:** $T_{\text{hot,out}} = 52^\circ\text{C}$ (system alert)
- **Alert level:** $T_{\text{hot,out}} = 50^\circ\text{C}$ (increase cooling water flow)
- **Emergency:** $T_{\text{hot,out}} < 48^\circ\text{C}$ (automatic dilution + shutdown)

Dilution System Design

- Fresh water reservoir located near absorber
- Solenoid valve triggered when $T < 50^\circ\text{C}$
- Flow rate: 2–5 L/min for a 10 kW system
- Operates until $T > 55^\circ\text{C}$ is restored

7.3 Comparison with Alternatives

Economic and Technical Comparison[1],[2],[3].

Technology	COP	Capital Cost (€/kW)	Best Application
Vapor Compression	3.0	2,500	Grid-connected, reliable power
Single-Effect Absorption	0.8	1,100	Solar thermal, waste heat
Double-Effect Absorption	1.2	1,300	High-temp solar, industrial
Adsorption	0.6	1,400	Remote, minimal maintenance
Thermoelectric	2.0	3,300	Niche; limited capacity

**Operating cost evaluated over 10-year horizon.*[2],[3]

Break-Even Analysis (Single vs. Vapor Compression)

Single-effect absorption becomes economically viable when:

$$C_{\text{heat}} \times \left(\frac{Q_{\text{gen}}}{COP_{\text{VC}}} \right) < C_{\text{elec}} \times W_{\text{VC}}$$

For tropical regions with $C_{\text{elec}} = 0.10/\text{kWh}$ and $C_{\text{heat}} = 0.02/\text{kWh}$:

- Annual cooling demand: 50 MWh [1]
- Vapor-compression operating cost: €15,000/year [1]
- Absorption operating cost: €4,600/year
- **Annual savings:** €10,400 → [1] **Payback period:** 4–5 years [1]

8 CONCLUSION

8.1 Key Findings

Summary of Parametric Investigation

A comprehensive parametric analysis examined nine operating parameters influencing the single-effect LiBr–H₂O absorption cooling system. Results indicate the following sensitivity hierarchy:

High-Sensitivity Parameters (>15% COP variation)

- Strong solution concentration (x_{strong}): 53.4% variation
- Weak solution concentration (x_{weak}): 31.5% variation
- Generator temperature (T_G): 22.6% variation
- Solution heat exchanger effectiveness (ε_{SHX}): 19.0% variation
- Absorber temperature (T_A): 17.1% variation

Moderate-Sensitivity Parameter

- System losses ($loss_{\text{frac}}$): 4.8% COP penalty across 1–5% range, exhibiting linear dependence

Low-Sensitivity Parameters (<1% COP variation)

- Condenser temperature (T_C)
- Evaporator temperature (T_E)
- Pump efficiency (η_{pump})

These results highlight that performance enhancement efforts should prioritize **solution concentrations, generator temperature, and heat exchanger effectiveness**, while low-sensitivity parameters offer **design flexibility** without significant thermodynamic impact.

8.2 Optimized System Performance

Integrated Optimization Summary

Combined optimization of high-sensitivity parameters results in an overall COP enhancement of **31.9%**, with corresponding reductions in generator load and collector area.

Metric	Baseline	Optimized	Change
COP	0.721	0.950	+31.9%
Generator Heat (kW)	14.93	11.34	–24.0%
Collector Area (m ²)	38	29	–23.7%

Key Contributions: x_{strong} (+18.6%), x_{weak} (+15.7%), T_G (+13.1%), ε_{SHX} (+9.5%), T_A (+7.8%), and $loss_{\text{frac}}$ (+1.4%) collectively contribute to the overall performance gain.

8.3 Design Recommendations

Parameter Optimization Sequence

1. Optimize x_{strong} to 0.65 with crystallization monitoring.
2. Reduce x_{weak} to 0.48–0.50 through absorber design enhancement.
3. Target generator temperature $T_G = 80^\circ\text{C}$ via solar collector selection.
4. Enhance solution heat exchanger effectiveness ε_{SHX} to 0.85 using improved design.
5. Increase absorber temperature T_A to 36–40°C by improving cooling system efficiency.
6. Minimize system losses ($\text{loss}_{\text{frac}} < 2\%$) with advanced thermal insulation.

Non-critical parameters — T_C , T_E , and η_{pump} — may be chosen based on application type and economic considerations.

8.4 Limitations and Future Work

Model Constraints: Steady-state analysis only; transient effects such as solar intermittency require dynamic simulation. Assumes ideal mass and energy balances, while real systems may deviate by 3–6%. Crystallization limits are empirically bounded; full phase equilibrium modeling could refine these limits. Control system dynamics are not included, so operational lag and oscillations are not captured.

Validity Range:

- Generator temperature: 75–100°C (solar thermal domain)
- Evaporator duty: 10–50 kW (small-to-medium systems)
- Ambient conditions: 20–50°C (tropical/subtropical focus)
- System pressure: Sub-atmospheric (>100 kPa high-pressure side)

Future Research Directions

Future work can focus on enhancing performance, applicability, and robustness of absorption cooling systems:

- **Experimental Validation:** Laboratory testing across the full operating envelope to confirm parametric trends and model predictions[1].
- **Multi-Effect Integration:** Development of double-effect systems for high-temperature solar thermal applications[2].
- **Hybrid Systems:** Coupling absorption with vapor-compression cycles for improved part-load efficiency and operational flexibility[1].
- **Advanced Sorbents:** Exploration of alternative working pairs such as $\text{NH}_3\text{--H}_2\text{O}$ or deep eutectic solvents for enhanced thermodynamic performance[2].
- **Dynamic Simulation:** Use of TRNSYS or similar tools to capture transient behavior under variable solar insolation[1].
- **Waste Heat Recovery:** Integration with industrial process heat sources to improve overall energy utilization[1].

8.5 Sustainability and Environmental Impact

Solar-powered single-effect absorption cooling in tropical regions offers significant environmental and resource advantages compared to grid-connected vapor-compression systems:

- **Energy Savings:** 40–50% annual reduction in electricity consumption.
- **GHG Reduction:** 2–3 tonnes CO₂ emissions avoided per kW per year.
- **Water Conservation:** Up to 60% lower water usage due to air-cooled operation.
- **Longevity:** 20–25 year system lifetime, exceeding typical vapor-compression systems.
- **Environmental Safety:** Zero ozone depletion potential; LiBr–H₂O is non-toxic and CFC-free.

Strategic Applications: Off-grid rural healthcare (vaccine/medicine storage), agricultural processing (cold chain), remote research stations, tropical residential cooling, and data centers leveraging waste heat recovery.

8.6 Final Conclusions

Single-effect LiBr–H₂O absorption cooling systems are a mature and thermodynamically robust technology for solar thermal applications in tropical and subtropical regions. Systematic parametric optimization demonstrates a potential **31.9% COP improvement** through targeted manipulation of high-sensitivity parameters:

- **Primary drivers:** Strong solution concentration ($x_{\text{strong}} = 0.65$) and weak solution concentration ($x_{\text{weak}} = 0.50$), contributing up to 30% combined improvement.
- **Secondary parameters:** Generator temperature, absorber temperature, and SHX effectiveness, contributing 7–13% individually.
- **Tertiary factors:** System losses minimized via premium insulation and component selection, providing additional efficiency gains.

This sensitivity-based framework enables engineers to prioritize design efforts for maximum thermodynamic benefit, offering cost-effective solutions in resource-constrained environments. As global cooling demand rises and renewable energy adoption becomes increasingly critical, solar-powered absorption cooling systems provide a sustainable and reliable pathway for thermal energy management in developing regions with abundant solar resources and limited electrical infrastructure.

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APPENDIX A: PARAMETRIC STUDY RESULTS FROM EES SIMULATIONS

A.1 Generator Temperature (T_G) vs COP

Table 2: EES Parametric Results: Generator Temperature Effect (Source: tg vs cop.pdf)

Run	T_G (°C)	COP _{with,pump}	Change (%)
1	75	0.8149	+13.1
2	77.78	0.7966	+10.5
3	80.56	0.7787	+8.0
4	83.33	0.7612	+5.6
5	86.11	0.7442	+3.2
6	88.89	0.7275	+0.9
7	91.67	0.7113	-1.3
8	94.44	0.6955	-3.5
9	97.22	0.6801	-5.7
10	100	0.6650	-7.8
Total Variation		22.6%	Inverse

A.2 Weak Solution Concentration (x_{weak}) vs COP

Table 3: EES Parametric Results: Weak Solution Concentration Effect (Source: x_weak vs cop.pdf)

Run	x_{weak}	COP _{with,pump}	Change (%)
1	0.48	0.8342	+16.9
2	0.4889	0.8252	+15.7
3	0.4978	0.8143	+13.9
4	0.5067	0.8010	+11.3
5	0.5156	0.7849	+8.9
6	0.5244	0.7655	+6.2
7	0.5333	0.7420	+3.0
8	0.5422	0.7133	-1.1
9	0.5511	0.6781	-5.8
10	0.56	0.6346	-11.9
Total Variation		31.5%	Inverse

A.3 Strong Solution Concentration (x_{strong}) vs COP

Table 4: EES Parametric Results: Strong Solution Concentration Effect (Source: x_strong vs cop.pdf)

Run	x_{strong}	COP _{with,pump}	Change (%)
1	0.58	0.5576	-22.6
2	0.5878	0.6090	-15.6
3	0.5956	0.6531	-9.4
4	0.6033	0.6916	-4.1
5	0.6111	0.7256	+0.6
6	0.6189	0.7562	+4.9
7	0.6267	0.7840	+8.7
8	0.6344	0.8095	+12.3
9	0.6422	0.8331	+15.6
10	0.65	0.8551	+18.6
Total Variation		53.4%	Direct

A.4 Solution Heat Exchanger Effectiveness (ε_{SHX}) vs COP

Table 5: EES Parametric Results: SHX Effectiveness Effect (Source: e_shx vs cop.pdf)

Run	ε_{SHX}	COP _{with,pump}	Change (%)
1	0.50	0.6634	-8.0
2	0.5444	0.6753	-6.3
3	0.5889	0.6878	-4.6
4	0.6333	0.7007	-2.8
5	0.6778	0.7141	-1.0
6	0.7222	0.7280	+0.8
7	0.7667	0.7425	+3.0
8	0.8111	0.7576	+5.1
9	0.8556	0.7732	+7.2
10	0.90	0.7896	+9.5
Total Variation		19.0%	Direct

A.5 System Loss Fraction ($\text{loss}_{\text{frac}}$) vs COP

Table 6: EES Parametric Results: System Loss Effect (Source: loss_frac vs cop.pdf)

Run	Loss Fraction (%)	COP _{with,pump}	Change (%)
1	0.00	0.7354	+2.0
2	0.5556	0.7313	+1.4
3	1.111	0.7273	+0.9
4	1.667	0.7234	+0.3
5	2.222	0.7194	-0.2
6	2.778	0.7155	-0.8
7	3.333	0.7117	-1.3
8	3.889	0.7079	-1.8
9	4.444	0.7041	-2.4
10	5.00	0.7004	-2.9
Total Variation		-4.76%	Linear

A.6 Absorber Temperature (T_A) vs COP

Table 7: EES Parametric Results: Absorber Temperature Effect (Source: ta vs cop.pdf)

Run	T_A (°C)	COP _{with,pump}	Change (%)
1	25	0.6644	-7.8
2	26.67	0.6764	-6.2
3	28.33	0.6885	-4.5
4	30	0.7008	-2.8
5	31.67	0.7132	-1.1
6	33.33	0.7258	+0.7
7	35	0.7385	+2.4
8	36.67	0.7514	+4.2
9	38.33	0.7644	+6.0
10	40	0.7776	+7.8
Total Variation		17.06%	Direct

A.7 Condenser Temperature (T_C) vs COP

Table 8: EES Parametric Results: Condenser Temperature Effect (Source: tc vs cop.pdf)

Run	T_C ($^{\circ}\text{C}$)	$\text{COP}_{\text{with,pump}}$	Change (%)
1	30	0.7159	-0.7
2	31.67	0.7167	-0.6
3	33.33	0.7175	-0.5
4	35	0.7184	-0.4
5	36.67	0.7192	-0.2
6	38.33	0.7200	-0.1
7	40	0.7209	+0.0
8	41.67	0.7217	+0.1
9	43.33	0.7226	+0.2
10	45	0.7234	+0.3
Total Variation		+1.05%	Negligible

A.8 Evaporator Temperature (T_E) vs COP

Table 9: EES Parametric Results: Evaporator Temperature Effect (Source: te vs cop.pdf)

Run	T_E ($^{\circ}\text{C}$)	$\text{COP}_{\text{with,pump}}$	Change (%)
1	0	0.7219	+0.1
2	1.111	0.7211	-0.0
3	2.222	0.7204	-0.1
4	3.333	0.7196	-0.2
5	4.444	0.7188	-0.3
6	5.556	0.7181	-0.4
7	6.667	0.7173	-0.5
8	7.778	0.7166	-0.6
9	8.889	0.7158	-0.7
10	10	0.7151	-0.8
Total Variation		-0.94%	Negligible

A.9 Pump Efficiency (η_{pump}) vs COP

Table 10: EES Parametric Results: Pump Efficiency Effect (Source: pump_eff vs cop.pdf)

Run	η_{pump}	COP _{with,pump}	Change (%)
1	0.60	0.7209	-0.01
2	0.6333	0.7210	-0.00
3	0.6667	0.7210	-0.00
4	0.70	0.7210	-0.00
5	0.7333	0.7210	-0.00
6	0.7667	0.7210	-0.00
7	0.80	0.7210	+0.00
8	0.8333	0.7210	+0.00
9	0.8667	0.7210	+0.00
10	0.90	0.7210	+0.00
Total Variation		0.02%	Negligible

*Note: All parametric tables extracted from EES version 9.172 simulation outputs.
Baseline configuration: $T_G = 90C$, $T_C = 40.2C$, $T_A = 32.7C$, $T_E = 1.3C$, $x_{strong} = 0.61$, $x_{weak} = 0.54$, $\varepsilon_{SHX} = 0.70$, $\eta_{pump} = 0.75$, $loss_{frac} = 0.02$.*

APPENDIX B: COMPONENT RATING SHEETS AND SPECIFICATIONS

This appendix presents detailed component specifications, rating sheets, and design parameters referenced in the optimization strategy and system design recommendations.

C.1 Solution Heat Exchanger (SHX) Specifications

Baseline SHX Configuration ($\varepsilon_{SHX} = 0.70$)

Table 11: Baseline Solution Heat Exchanger Specifications

Specification	Value	Unit	Source
Effectiveness (ε_{SHX})	0.70	–	Design baseline
Hot-side mass flow	0.92	kg/s	EES Model [6]
Cold-side mass flow	1.00	kg/s	EES Model [6]
Hot inlet temperature	53.3	°C	Calculated
Cold inlet temperature	32.7	°C	Baseline condition
Heat transfer rate	22.5	kW	Calculated
Fouling factor	0.0001	m ² K/W	Standard [6]
Material	Copper tubes	–	Industry standard
Tube diameter	19	mm	Standard [6]
Tube length	1.2	m	Design
Number of tubes	45	–	Design
Pressure drop (hot side)	3.2	kPa	Calculated
Pressure drop (cold side)	2.8	kPa	Calculated

Enhanced SHX Configuration ($\varepsilon_{SHX} = 0.85$) – Optimization Target

Table 12: Enhanced Solution Heat Exchanger Specifications (Optimized)

Specification	Value	Unit	Improvement
Effectiveness (ε_{SHX})	0.85	–	+21.4%
Surface area increase	+35%	–	Design requirement
Tube diameter (enhanced)	16	mm	Reduced for higher h
Number of tubes	62	–	+38%
Tube-side velocity	1.5	m/s	Optimized
Shell-side velocity	0.8	m/s	Optimized
Pressure drop (max)	5.0	kPa	Acceptable limit
Heat transfer rate	25.8	kW	+14.7%
Material	Copper-Nickel alloy	–	Fouling resistance
Capital cost increase	+18–22%	–	vs. baseline
Payback period	2.5–3.2	years	10 kW system

Note: Enhanced SHX design justified by 9.5% COP improvement potential over baseline configuration. Copper-Nickel construction recommended for LiBr solution

compatibility and anti-fouling properties.

C.2 Absorption Chiller Nominal Ratings

Single-Effect LiBr-H₂O Chiller (Reference Configuration)

Table 13: Single-Effect Absorption Chiller Nominal Ratings [2]

Parameter	Specification	Unit	Source
Cooling Performance			
Cooling capacity (nominal)	10–350	kW	[2]
COP (rated conditions)	0.73–0.79	–	[2]
COP (optimized model)	0.95	–	This work
Operating Temperatures			
Generator inlet (hot source)	90	°C	Baseline
Generator inlet (optimized)	80	°C	Target [This work]
Condenser inlet temp	40.2	°C	Baseline
Evaporator outlet (chilled water)	1.3	°C	Baseline
Absorber temperature	32.7	°C	Baseline
Working Fluid Properties			
Refrigerant	Water	–	[2]
Absorbent	Lithium Bromide	–	[2]
Strong solution concentration	0.61	–	Baseline
Strong solution (optimized)	0.65	–	Target [This work]
Weak solution concentration	0.54	–	Baseline
Weak solution (optimized)	0.48	–	Target [This work]
Crystallization limit	42	°C	[2]
Safe operating margin	50	°C	Design safety [This work]
Physical Specifications			
Footprint	2.5 × 1.8	m	Typical [2]
Weight (dry)	800	kg	Typical
Design pressure	0.5–2.0	MPa	[2]
Material (tubes)	Copper	–	[2]
Material (shell)	Carbon steel	–	[2]

Double-Effect LiBr-H₂O Chiller (Commercial Reference) [3]

Table 14: Double-Effect Absorption Chiller Commercial Specifications [3]

Parameter	Specification	Unit	Source
Cooling Performance			
Cooling capacity	116–1160	kW	[3]
COP (rated)	1.35–1.41	–	[3]
Improvement vs. single-effect	+60–70	%	[2]
Operating Temperatures			
High-pressure generator inlet	180	°C	[3]
Low-pressure generator inlet	110–120	°C	[3]
Condenser inlet	30	°C	[3]
Chilled water outlet	7	°C	[3]
Solution Properties			
Strong solution concentration	56–57	% LiBr	[3]
Weak solution concentration	50–52	% LiBr	[3]
Crystallization limit	42	°C	[3]
System Requirements			
Heat source type	Solar/industrial	–	[3]
Solar collector type	Evacuated tube	–	[9]
Collector area per kW cooling	3.5–4.5	m ² /kW	[9]

C.3 Solar Collector Performance Data

Flat-Plate Collector (FPC) Configuration

Table 15: Flat-Plate Solar Collector Specifications [9]

Parameter	Value	Unit	Source
Peak efficiency	60–65	%	[9]
Outlet temperature range	50–65	°C	[9]
Operating pressure	3–8	bar	[9]
Heat loss coefficient (U_L)	7.5–10	W/m ² K	[9]
Aperture area (typical)	2–10	m ²	[9]
Absorber material	Copper	–	[9]
Cover material	Tempered glass	–	[9]
Weight per unit area	60–80	kg/m ²	[9]
Application			
Suitability for single-effect	Excellent	–	[9]
System scale	Small–medium	–	[9]
Installation cost	€400–600/m ²	–	[9]

Evacuated Tube Collector (ETC) Configuration

Table 16: Evacuated Tube Solar Collector Specifications [9]

Parameter	Value	Unit	Source
Peak efficiency	75–80	%	[9]
Outlet temperature range	80–100	°C	[9]
Operating pressure	6–15	bar	[9]
Heat loss coefficient (U_L)	1.5–2.5	W/m ² K	[9]
Tube diameter	47–58	mm	[9]
Tube length	1500–2000	mm	[9]
Number of tubes (typical)	20–30	–	[9]
Weight per unit area	25–35	kg/m ²	[9]
Application			
Suitability for single-effect	Very good	–	[9]
Suitability for double-effect	Excellent	–	[9]
Part-load performance	Superior	–	[9]
System scale	Medium–large	–	[9]
Installation cost	€500–800/m ²	–	[9]

Parabolic Trough Collector (PTC) Configuration [12]

Table 17: Parabolic Trough Solar Collector Specifications [12]

Parameter	Value	Unit	Source
Peak efficiency	70–75	%	[12]
Outlet temperature range	150–200	°C	[12]
Operating pressure	25–40	bar	[12]
Heat loss coefficient (U_L)	0.5–1.5	W/m ² K	[12]
Concentration ratio	80–100	–	[12]
Aperture width	5–6	m	[12]
Module length	100	m	[12]
Tracking system	1-axis solar	–	[12]
Receiver tube diameter	70	mm	[12]
Receiver material	Stainless steel	–	[12]
Heat transfer fluid	Therminol VP-1	–	[12]
Application			
Suitability for double-effect	Excellent	–	[12]
Suitability for triple-effect	Good	–	[12]
System scale	Large industrial	–	[12]
Installation cost	€800–1200/m ²	–	[12]
Land requirement	6–8	m ² /kW	[12]

Comparative Summary:

- **Single-effect LiBr-HO systems:** Optimal with flat-plate or evacuated-tube collectors (50–100°C). COP baseline 0.70–0.79, achievable 0.95 with optimization.

- **Double-effect systems:** Require evacuated-tube (80–100°C min) or parabolic trough collectors (150–200°C). COP range 1.22–1.42.
- **Collector selection criterion:** Match outlet temperature to generator inlet requirement; evacuated-tube recommended for tropical solar applications.

C.4 Component Integration Matrix

Table 18: Component Compatibility and Integration Requirements

Component	Single-Effect	Double-Effect	Material	Ref
Evaporator	Copper tubes	Copper tubes	Cu–Ni alloy	[2]
Absorber	Copper spray	Copper spray	Cu–Ni alloy	[2]
Generator (HTG)	Copper tubes	Copper tubes	Cu–Ni alloy	[3]
Low-pressure generator	–	Copper tubes	Cu–Ni alloy	[3]
Condenser	Copper tubes	Dual condensers	Cu–Ni alloy	[3]
SHX-I	Plate-frame	Plate-frame	Copper/brass	[6]
SHX-II	–	Plate-frame	Copper/brass	[3]
Solution pump	Centrifugal	Dual pumps	Cast iron	[2]
Expansion valve	Capillary/throttle	Dual valves	Brass	[2]
Piping	Copper tubing	Copper tubing	Cu (seamless)	[2]
Insulation	Foam/mineral wool	Foam/mineral wool	Standard	[This work]
Instrumentation	T, P sensors	T, P, X sensors	Stainless steel	[This work]

Key Material Notes:

- Copper-Nickel (Cu-Ni) alloys preferred for LiBr solution compatibility (inhibits corrosion and crystallization nucleation)
- Stainless steel (316L) recommended for sensors and valve bodies in high-concentration zones
- Thermal insulation (≥50 mm) mandatory for piping to minimize system losses below 2% target
- Solenoid valves for dilution system must be rated for corrosive LiBr environment

References cited in Appendix C:

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APPENDIX C: EES SOURCE CODE FOR SINGLE-EFFECT LiBr-H₂O ABSORPTION SYSTEM

```

"=====
" SOLAR ABSORPTION COOLING SYSTEM (LiBr-H2O Single-Effect) "
" Using Patek & Klomfar / ASHRAE validated correlations "
"=====

"----- SMALL UTILITY FUNCTION -----"
FUNCTION Flag_Tinv(T1, T2)
IF (T1 = -999) OR (T2 = -999) THEN
    Flag_Tinv = 1
ELSE
    Flag_Tinv = 0
ENDIF
END

"----- PROPERTY FUNCTIONS (published) -----"
FUNCTION Psat_water(T) "Antoine equation for pure water [kPa]"
A = 8.07131
B = 1730.63
C = 233.426
Psat_water = 0.133322 * 10^(A - B/(C + T))
END

FUNCTION h_LiBr(T, X) "Specific enthalpy [kJ/kg] (Patek & Klomfar simplified)"
" T in C, X in mass fraction (0 1 )"
h_LiBr = -408.2 + 4.67*T - 0.0173*T^2 + 1988*X - 1970*X^2 + 0.404*T*X
END

FUNCTION Cp_LiBr(T, X) "Specific heat [kJ/kg-K] ( h / T )"
Cp_LiBr = 4.67 - 0.0346*T + 0.404*X
END

FUNCTION rho_LiBr(T, X) "Density [kg/m3] (Gomri / ASHRAE fit)"
rho_LiBr = (1000 - 0.3*(T - 20)) + 750*X
END

"----- INVERSION: Temperature from enthalpy -----"
FUNCTION T_from_h_LiBr(h, X)
a = -0.0173
b = 4.67 + 0.404*X
c = -408.2 + 1988*X - 1970*X^2 - h
disc = b^2 - 4*a*c
IF disc < 0 THEN
    T_from_h_LiBr = -999
ELSE
    T_from_h_LiBr = (-b + SQRT(disc)) / (2*a)
ENDIF
END

"----- BASE INPUTS -----"
Q_evap = 10.77 {kW}
T_G = {C}
T_C = {C}
T_A = {C}
T_E = {C}
x_strong =
x_weak =
eps_SHX =
pump_eff =
loss_frac =

"----- REFRIGERANT (WATER) PROPERTIES -----"
h_evap_vap = ENTHALPY(Water, T=T_E, x=1)
h_evap_liq = ENTHALPY(Water, T=T_E, x=0)
h_cond_vap = ENTHALPY(Water, T=T_C, x=1)
h_cond_liq = ENTHALPY(Water, T=T_C, x=0)
h_fg_evap = h_evap_vap - h_evap_liq
h_fg_cond = h_cond_vap - h_cond_liq

"----- MASS FLOWS -----"
m_r = Q_evap / h_fg_evap {kg/s}
m_sol = m_r / (x_strong - x_weak) {kg/s}

"----- SHX (enthalpy-based, corrected) -----"
"Hot stream: strong solution leaving generator (T_G)"
h_hot_in = h_LiBr(T_G, x_strong)
cp_hot = Cp_LiBr(T_G, x_strong)

"Cold stream: weak solution leaving absorber (T_A)"
h_cold_in = h_LiBr(T_A, x_weak)
cp_cold = Cp_LiBr(T_A, x_weak)
C_hot = m_sol * cp_hot
C_cold = m_sol * cp_cold
Q_max = MIN(C_hot, C_cold) * (T_G - T_A)
Q_SHX = eps_SHX * Q_max

"Enthalpy balances"
h_hot_out = h_hot_in - Q_SHX / m_sol
h_cold_out = h_cold_in + Q_SHX / m_sol

"Temperature from enthalpy"
T_hot_out_raw = T_from_h_LiBr(h_hot_out, x_strong)
T_cold_out_raw = T_from_h_LiBr(h_cold_out, x_weak)

```

```

Flag_T_inversion = Flag_Tinv(T_hot_out_raw, T_cold_out_raw)

"Enforce physical bounds"
T_hot_out = MAX(T_A, MIN(T_hot_out_raw, T_G))
T_cold_out = MAX(T_A, MIN(T_cold_out_raw, T_G))

"----- GENERATOR/CONDENSER (enthalpy-based) -----"
"CRITICAL CHANGE: Use actual enthalpy from SHX outlet"
"Refrigerant desuperheat/condense"
Q_des = m_r * h_fg_cond

"Sensible heat in generator"
h_gen_inlet = h_cold_out
h_gen_outlet = h_LiBr(T_G, x_strong)
Q_sens = m_sol * (h_gen_outlet - h_gen_inlet)

"Total generator heat"
Q_gen = (Q_des + Q_sens) * (1 + loss_frac)

"----- PUMP WORK -----"
rho_sol = rho_LiBr((T_A + T_G)/2, (x_weak + x_strong)/2)
P_low = Psat_water(T_A)
P_high = Psat_water(T_G)
deltaP = (P_high - P_low) * 1000
Wp = m_sol * deltaP / (rho_sol * pump_eff) / 1000

"----- PERFORMANCE -----"
COP_no_pump = Q_evap / Q_gen
COP_with_pump = Q_evap / (Q_gen + Wp)

"----- CRYSTALLIZATION CHECK -----"
T_cryst_limit = 35
Warning_Crystallization = MAX(0, SIGN(T_cryst_limit - T_hot_out))

"----- OUTPUTS -----"
eps_SHX_out = eps_SHX
m_r_out = m_r
m_sol_out = m_sol
Q_SHX_out = Q_SHX
T_hot_out_out = T_hot_out
T_cold_out_out = T_cold_out
T_cold_in_out = T_cold_out
h_hot_in_out = h_hot_in
h_hot_out_out = h_hot_out
h_cold_in_out = h_cold_in
h_cold_out_out = h_cold_out
h_gen_inlet_out = h_gen_inlet
Q_SHX_out_check = Q_SHX
Q_des_out = Q_des
Q_sens_out = Q_sens
Q_gen_out = Q_gen
rho_sol_out = rho_sol
Wp_out = Wp
COP_no_pump_out = COP_no_pump
COP_with_pump_out = COP_with_pump
Inversion_Flag_out = Flag_T_inversion
Warning_Crystallization_out = Warning_Crystallization

```

Listing 1: Complete EES Model Code